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TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN

RESEARCH REPORT
Digital Electronics

TOPIC:

Design a Traffic Light Controller at Crossroad

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SUMMARY

Digital electronics is a crucial field in electronic engineering, focusing on the processing and control of digital signals. In this project, our team will focus on the design and implementation of a traffic light controller at an intersection to efficiently and safely manage traffic flow.

This report is divided into the following main chapters:

Chapter I: Introduction - Introduces the reasons and objectives for researching the traffic light controller in the context of modern traffic.

Chapter II: Theoretical Background - Presents the fundamental theory of digital electronic circuits, including Flip-Flops, counters, decoders, and control algorithms.

Chapter III: Circuit Design and Simulation - Provides a detailed guide on constructing and simulating the traffic light controller using electronic simulation tools. The steps include circuit diagram design, operational simulation, and performance optimization.

Chapter IV: Experimental Results - Presents the experimental results, including traffic light states, switching times between lights, and the evaluation of the control system's effectiveness.

Chapter V: Conclusion and Future Directions - Summarizes the achievements of the project and proposes future directions to enhance the system's effectiveness and features.

The team hopes that this project will provide an effective solution for traffic control at intersections and open new avenues for development in the field of digital electronics and automated control.

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Glossary

State	Light 1	Light 2	Alias
1	Red	Green	RG
2	Red	Yellow	RY
3	Yellow	Red	YR
4	Green	Red	GR

CHAPTER I: INTRODUCTION

1. Problem Statement

In the context of increasingly complex traffic, the management and control of traffic flow at intersections have become a significant challenge. Traffic lights not only play a role in regulating traffic but also directly impact the safety and efficiency of vehicle movement.

Currently, some cities have experimented with removing the countdown timers on traffic lights to reduce the tendency of drivers to rush through intersections when the light changes from green to red. However, this solution has faced significant public criticism due to the uncertainty it creates and the difficulty for drivers in estimating when to stop or proceed. The removal of countdown timers replaces a clear and transparent method with an approach that is less decisive, potentially causing confusion for drivers, especially in complex traffic environments like those in Vietnam.

2. Rationale for the Study

Given these challenges, it is necessary to research and develop a traffic light control system at intersections that uses LEDs to display green, red, and yellow signals without showing countdown timers. Such a system not only needs to ensure traffic safety but also must be easily understood and accepted by road users.

3.Research Objectives

The objective of this study is to design and implement a traffic light control system at intersections using LEDs without the need for countdown timers. This system must ensure:

- The reasonable allocation of signal times based on traffic volume and time of day.
- The safety and transparency in traffic regulation.
- Ease of operation and maintenance.

4.Scope of the Study

The study will focus on designing a traffic light control system for simple intersections. More complex elements, such as intelligent control systems or the integration of artificial intelligence for flexible adjustment, will be considered in future research.

5. Introduction The traffic light

The traffic light controller circuit at a roundabout with four entry/exit points is a highly applicable project that offers numerous benefits for managing and improving urban traffic. It meets requirements such as automation, high reliability, and energy efficiency.

CHAPTER II: THEORETICAL BASIS

1. **Counting Block:** Uses D flip-flops capable of performing both up-counting and down-counting. In the project, the down-counting feature of D flip-flops is applied to count values in the units and tens digits.

2. **Logic Gates:** Used primarily for setting or clearing data. They have a significant impact on the accurate operation of the entire circuit.

3. LEDs:

Quiet Junction

Red	Green
Red	Amber
Green	Red
Amber	Red

Busy Junction

Red	Green
Red	Amber
Red	Red
Green	Red
Amber	Red
Red	Red

Assumed Cycling Rules

One Light	Cross Light	Comments
Green	Red	Traffic moving on one street
Yellow	Red	Traffic on cross street must wait for this light to turn red.
Red	Red	Both lights are red for about one second.
Red	Green	Cross traffic now moves.

This is the basic sequence for a traffic light without turn signals or features such as an “advanced green”, etc.

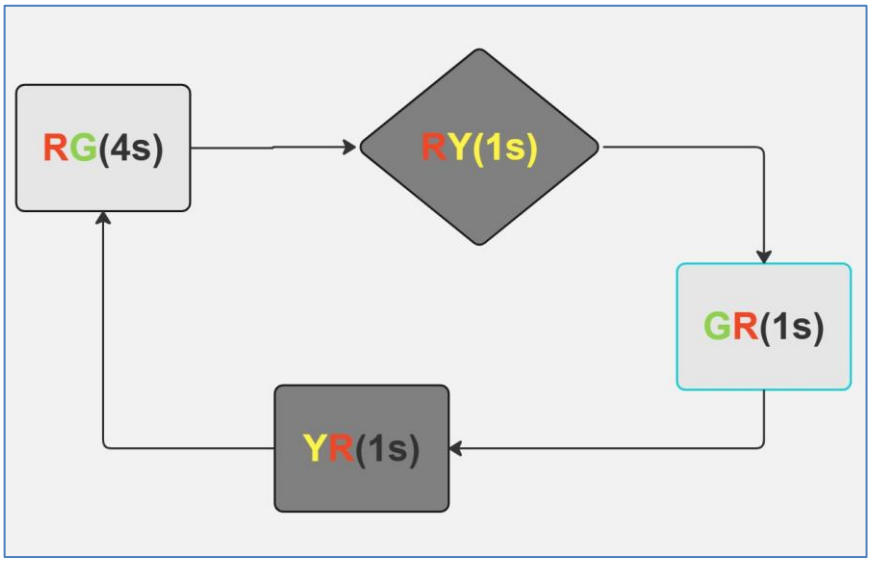
The traffic lights indicate whether vehicles are allowed to move or must stop, corresponding to green and red signals.

The forward biasing for the signal lights is controlled by the output of the D flip-flop (FF D).

CHAPTER III: DESIGN, ANALYSIS, AND IMPLEMENTATION OF THE PROGRAM

1) DESIGN

1.1 Diagram for the System



1.2 Assign state numbers to flip-flop outputs and construct the transition table.

D	Clk	Q_n	Q_{n+1}
X	0	0	0
X	0	1	1
0	1	X	0
1	1	X	1

Truth Table for D Latch

State	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	R1	Y1	G1	R2	Y2	G2
1	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1
2	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	1
3	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	1
4	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	1
5	0	0	0	0	1	0	0	0	0	0	1	0	0	0	1	0
6	0	0	0	0	0	1	0	0	0	0	0	0	1	1	0	0
7	0	0	0	0	0	0	1	0	0	0	0	0	1	1	0	0
8	0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	0
9	0	0	0	0	0	0	0	0	1	0	0	0	1	1	0	0
10	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	0

	Present state										Next state									
	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
2	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
3	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
4	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
5	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
6	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0
7	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0
8	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0
9	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0
10	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0

Counting Circuit for the Units Place

Constructed using 10 D flip-flops (FFs). The circuit functions to count up and read unit values ranging from 0 to 9, then repeats the cycle, returning to 0. Whenever a clock pulse is present, all values are processed. Since there is no prohibited state, the D flip-flops will always output a specific binary value regardless of the input.

Additionally, the 10 D flip-flops are connected to operate synchronously with each other, leading to faster processing speed and saving operational cycles.

Fill in K-Maps and determine the minimised expressions

Q0Q4 \ Q5Q9	00	01	11	10
00	0	0	X	0
01	1	X	X	X
11	X	X	X	X
10	1	X	X	X

R1=Q0+Q4

Q0Q4 \ Q5Q9	00	01	11	10
00	0	1	X	0
01	0	X	X	X
11	X	X	X	X
10	0	X	X	X

Y1 = Q9

Q0Q4 \ Q5Q9	00	01	11	10
00	0	0	X	1
01	0	X	X	X
11	X	X	X	X
10	0	X	X	X

G1 = Q5

Q0Q4 \ Q5Q9	00	01	11	10
00	0	1	X	1
01	0	X	X	X
11	X	X	X	X
10	0	X	X	X

R2 = Q5 +Q9

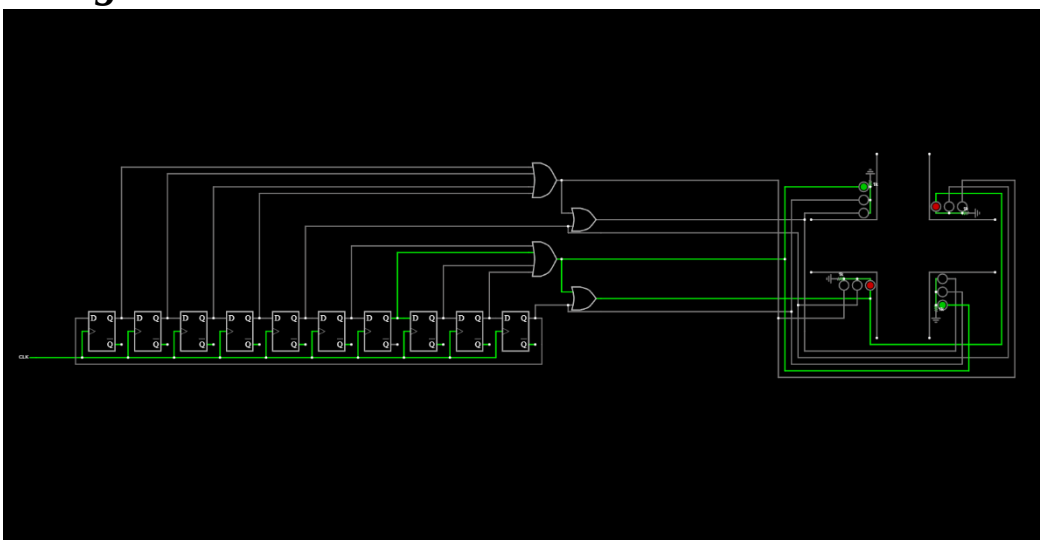
Q0Q4 \ Q5Q9	00	01	11	10
00	0	0	X	0
01	1	X	X	X
11	X	X	X	X
10	0	X	X	X

Y2 = Q4

Q0Q4 \ Q5Q9	00	01	11	10
00	0	0	X	0
01	0	X	X	X
11	X	X	X	X
10	1	X	X	X

G2 = Q0

1.3 Logic circuits:



2 Analysis:

The traffic light control circuit at the link you provided uses Flip-Flops (FF) and other logic components to control the traffic light signals (green, red, yellow) in a specific cycle. The operation principle of the circuit is as follows:

2.1 Circuit Structure:

- The circuit uses a total of 10 Flip-Flops to control the traffic light signals.
- The first 4 FFs control Green Light 1, and then this green light controls Red Light 2.
- The next FF controls Yellow Light 1 and Red Light 2.
- The next 4 FFs control Green Light 2 and Red Light 1.
- The final FF controls Yellow Light 2 and Red Light 1.

2.2 Operation Principle:

- Green Light 1: Controlled by the first 4 FFs. When these FFs are active, Green Light 1 will illuminate, allowing vehicles in the corresponding lane to move. The signal from Green Light 1 then controls Red Light 2, causing Red Light 2 to turn on.
- Yellow Light 1: After Green Light 1 turns off, the next FF will control Yellow Light 1, which lights up to warn that the signal is about to turn red. Simultaneously, Yellow Light 1 also controls Red Light 2.
- Green Light 2: After the cycle of Red Light 2 ends, the next 4 FFs will control Green Light 2, allowing vehicles from another lane to move. Green Light 2 then controls Red Light 1, causing Red Light 1 to turn on.

- Yellow Light 2: Similar to Yellow Light 1, Yellow Light 2 will illuminate after Green Light 2 turns off. The final FF controls Yellow Light 2, providing a warning of the signal change, and controls Red Light 1.

2.3 Control Cycle:

This circuit operates according to a preset cycle, continuously switching between light signals to ensure traffic is managed safely and efficiently.

2.4 K-map and Optimization:

- Since Lights 3 and 4 are connected in parallel with Lights 1 and 2, and the logic states of Q0, Q1, Q2, Q3, and Q5, Q6, Q7, Q8 are grouped when creating the K-map, this helps optimize the logic functions, reducing the number of logic gates needed.

In summary, this circuit uses FFs to alternately control the traffic light signals at an intersection, based on a pre-programmed cycle, ensuring safe and efficient traffic flow.

CHAPTER IV: RESULTS – EVALUATION

Video:

Screen Recording
2024-08-10 221215.r

Link: https://tinyurl.com/2d7hdm8k

Traffic Light Controller at the Intersection: Evaluation and Development Directions

The traffic light controller at the intersection has effectively managed traffic flow. Specifically, the system has successfully performed the function of controlling the lights according to preset cycles, ensuring the safe and orderly movement of vehicles. However, during testing and experimentation, several factors affecting system performance were identified:

1. **Overall Performance:**
   - The system accurately controlled the timing of the transitions between traffic lights, ensuring that the signals changed at the right time, thereby minimizing unnecessary waiting times at intersections.
   - The system operated well at intersections with moderate traffic flow; however, its performance may decline when managing traffic at more complex intersections with high vehicle volumes.
2. **Potential Factors Affecting Performance:**
   - **Circuit Design and Component Quality:** The selection of components and circuit design can impact system stability. If the circuit is not optimally designed, it may lead to issues such as electronic interference or delays in signal transitions.
   - **Testing Conditions:** Environmental factors like electromagnetic interference or instability in the power supply can affect the performance of the control system.
   - **Optimization of Control Algorithms:** Currently, the control algorithm uses fixed cycles and has not been fully optimized for dynamically changing traffic situations.
3. **Resolution Approaches:**
   - **Improvement of Circuit Design:** Enhance the quality of circuit design and choose appropriate components to ensure system stability and reliability in real-world environments.
   - **Optimization of Algorithms:** Consider integrating adaptive control algorithms or predictive control to improve the system's responsiveness in complex traffic situations.
   - **Testing Condition Improvements:** Optimize environmental testing conditions to minimize the impact of electronic interference and other external factors.

Chapter V: Conclusion and Development Directions

1. **Conclusion:**
   The project has successfully designed and implemented a traffic light controller at an intersection, using modern programming and digital techniques to ensure accuracy and efficiency in managing traffic flow. The system has been tested and demonstrated stable operation, and it can be easily adjusted to suit actual traffic conditions. This traffic light controller helps to optimize vehicle waiting times, reduce congestion, and enhance traffic safety.
2. **Development Directions:**
   - **Performance Enhancement:** Continue researching and developing optimization algorithms to improve the controller's performance, especially in complex traffic situations that require real-time processing.
   - **Application to Other Intersections:** Expand this traffic light control system to other intersections, including more complex ones with multiple lanes and directions.
   - **Integration with Artificial Intelligence:** Explore the possibility of integrating artificial intelligence techniques so the system can automatically adjust light timings based on real traffic conditions, forecast, and respond flexibly to emergency situations.
   - **Development of Software Tools:** Build a more user-friendly interface and software tools to support the installation and adjustment of the traffic light control system in practice.
   - **Research New Control Methods:** Continue researching new control methods such as adaptive control or model-based predictive control to improve traffic management at large intersections.